

AI-Based Fault Detection in Communication Transmission Lines Using Electromagnetic Parameter Analysis

Marvin A. Guantero

Department of Electronics and Communications Engineering,
De La Salle University, Manila 1004, Philippines

Lawrence Y. Materum

Department of Electronics and Communications Engineering,
De La Salle University, Manila 1004, Philippines

Abstract—Reliable fault detection in communication transmission lines is crucial for modern infrastructure. This paper presents a simulation-driven method for classifying transmission line faults — normal, open, short, and mismatch — using physically grounded electromagnetic features. By leveraging key parameters such as the reflection coefficient, voltage standing wave ratio (VSWR), and material properties (complex permittivity, conductivity), we build a dataset that mimics real-world fault scenarios. A support vector machine (SVM) classifier, trained on these engineered features, achieves near-perfect classification accuracy and transparent decision boundaries. Our approach balances theoretical analysis and practical machine learning, offering a robust, interpretable pipeline for early diagnosis. Future directions include multi-frequency feature augmentation, uncertainty calibration, and deployment in hardware monitoring systems.

Index Terms—Transmission lines, fault detection, reflection coefficient, electromagnetic feature engineering, VSWR, permittivity, conductivity, support vector machine, machine learning, AI-based, signal integrity, communication cables.

I. INTRODUCTION

Transmission lines are vital for communication, power, and sensor networks. Faults such as opens, shorts, and mismatches degrade performance and reliability [1], [2], [3]. Classical strategies use direct impedance measurement and reflectometry [1], [3], but complex cable networks benefit from model-based diagnosis. Recent advances in electromagnetic simulation and machine learning enable deeper insights from feature extraction based on transmission line theory [4], [5]. Frequency-domain features like the complex reflection coefficient (Γ), VSWR, and engineered material parameters present robust, interpretable descriptors for fault diagnosis.

This work simulates a coaxial transmission line over four fault conditions—normal, open, short, mismatch—systematically extracting features with strong physical meaning [6], [2], [5]. A support vector machine (SVM) classifier is then trained to separate the

fault types with high accuracy [7], [8]. The study demonstrates how reflection and material feature engineering founded on physical theory provides reliable and interpretable classification, echoing best practices in both electromagnetic analysis and machine learning.

The MATLAB simulation used in this study has been made publicly available to ensure reproducibility and further research collaboration. The simulation code can be accessed at: <https://github.com/mguantero/ai-based-fault-detection>.

II. LITERATURE AND BIBLIOGRAPHY REVIEW

Classical transmission line theory establishes that when a wave travelling along a line encounters a discontinuity (change in load impedance), part of its energy is reflected and part transmitted [1], [3], [2]. The ratio of reflected to incident voltage phasors defines the complex reflection coefficient. The Smith chart provides a geometric mapping between normalized impedance and reflection coefficient, supporting intuitive analysis of matching and fault behavior [4], [1].

Dielectric loss, conductor loss, and mismatch affect the effective complex permittivity $\epsilon_r = \epsilon' - j \text{varepsilonpsilon}''$ and conductivity σ , which in turn influence attenuation (α) and phase (β) constants of the propagating wave [2], [1]. Material dissipation can be characterized through the loss tangent $\tan \delta = \epsilon''/\epsilon'$, and for low-loss dielectrics, an approximate linkage between imaginary permittivity and effective conductivity is used: $\sigma \approx \omega \epsilon_0 \epsilon''$.

Open and short circuit terminations are limiting cases producing nearly total reflections with phase differences: Open faults tend toward $\Gamma \approx +1\angle 0^\circ$ and short faults toward $\Gamma \approx +1\angle 180^\circ$ [1]. Moderate mismatches produce intermediate reflection magnitudes and arbitrary phases, depending on the magnitude of the load impedance and its reactive content.

Machine learning approaches, especially multi-class SVMs using kernel methods (here radial basis func-

tion), can effectively separate fault types when features capture physically distinct regimes [7]. Because the simulated parameter distributions are intentionally well separated (e.g., permittivity ranges, reflection magnitudes), high classification accuracy is expected.

III. GOVERNING EQUATIONS AND FORMULA SUMMARY

The simulation and subsequent analysis rely on the following equations:

Transmission Line and Reflection

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad (1)$$

$$z = \frac{Z_L}{Z_0}, \quad \Gamma = \frac{z - 1}{z + 1}, \quad z = \frac{1 + \Gamma}{1 - \Gamma}, \quad (2)$$

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad (3)$$

Material and Loss Relations

$$\varepsilon_r = \varepsilon' - j\varepsilon'', \quad (4)$$

$$\tan \delta = \frac{\varepsilon''}{\varepsilon'}, \quad (5)$$

$$\sigma \approx \omega \varepsilon_0 \varepsilon'', \quad (6)$$

$$\beta \approx \omega \sqrt{\mu_0 \varepsilon_0 \varepsilon'_{\text{eff}}}, \quad (7)$$

$$\gamma = \alpha + j\beta, \quad (8)$$

where γ is the complex propagation constant.

Impedance From Reflection (Used in Figure 4)

$$Z_L = Z_0 \frac{1 + \Gamma}{1 - \Gamma}. \quad (9)$$

Feature Transformations (Preprocessing)

$$x' = \log_{10}(x) \quad (\text{for skewed positive features}), \quad (10)$$

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma}. \quad (11)$$

SVM Decision Function (Conceptual)

For a multi-class SVM (one-vs-one ECOC strategy), binary decision functions have the form:

$$f(\mathbf{x}) = \sum_{i=1}^N \alpha_i y_i K(\mathbf{x}, \mathbf{x}_i) + b, \quad (12)$$

with radial basis kernel

$$K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2). \quad (13)$$

IV. DATASET GENERATION PROCESS

A. Simulation Overview

A coaxial line of length $L = 10$ m, characteristic impedance $Z_0 = 50 \Omega$, and operating frequency $f = 1$ GHz is simulated for four fault classes:

- Normal (Matched):** $Z_L \approx Z_0$ with small random perturbations ($\approx 1\%$) producing low $|\Gamma|$ and $\text{VSWR} \approx 1$ –1.05.
- Open Fault:** $Z_L \gg Z_0$ ($\sim 10^6 \Omega$), yielding $|\Gamma| \rightarrow 1$ and VSWR large (capped numerically).
- Short Fault:** $Z_L \ll Z_0$ ($\sim 0.1 \Omega$), $|\Gamma| \approx 1$ with phase $\approx \pi$.
- Mismatch:** $Z_L = Z_0 \cdot m$ where $m \in [2.5, 4.0]$ (random), generating intermediate reflection magnitudes 0.3–0.7.

Figure 1 shows stylized transmission line schematics highlighting typical impedance regimes and random fault positioning for non-normal classes.

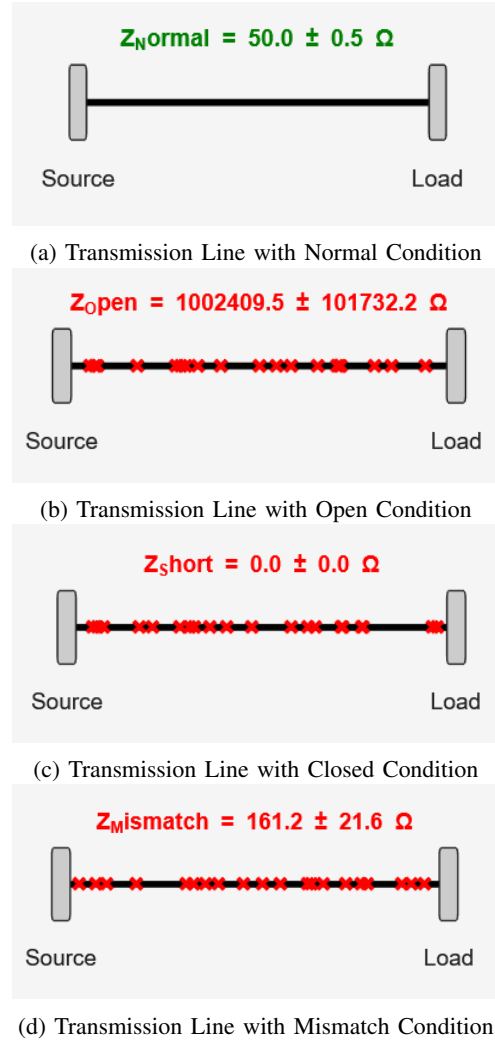


Fig. 1: Transmission line impedance schematic and representative spatial distribution of fault sites for Open, Short, and Mismatch cases. Normal conditions appear only at load termination with a near-matched impedance.

B. Representative Sample Rows

Table I provides one illustrative dataset row per fault type. Values adhere to the stochastic ranges defined in the simulator.

C. Physical Interpretation of Features

- A small $|\Gamma|$ with $VSWR \approx 1$ corresponds to efficient power transfer (matched condition).
- Large $|\Gamma|$ near unity indicates severe discontinuity (open or short) where almost all incident power reflects.
- Intermediate $|\Gamma|$ values correspond to mismatch faults where partial reflections degrade performance yet allow some power delivery.
- Imaginary permittivity and conductivity variations emulate dielectric and conductive loss changes induced by fault characteristics (e.g. arcing, corrosion, anomalous loading).

V. METHODOLOGY

A. Simulation of Transmission Line Faults

A 10 m coaxial line with a 50Ω characteristic impedance was modeled at 1 GHz, under four fault regimes:

- **Normal (Matched Load):** $Z_L \approx Z_0$; low random fluctuation reproduces manufacturing and environmental tolerance.
- **Open Fault:** $Z_L \gg Z_0$; a large resistance causes nearly total reflection.
- **Short Fault:** $Z_L \ll Z_0$; minimal resistance causes strong inverted reflection.
- **Mismatch:** Z_L randomly sampled in $[2.5Z_0, 4Z_0]$; intermediate reflection with variable phase.

Fault positions are randomized for non-Normal classes.

B. Electromagnetic Feature Engineering

Key features are calculated for every simulated sample:

- **Reflection Coefficient (Γ):** $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$, capturing mismatch severity and fault type via magnitude and phase.
- **Voltage Standing Wave Ratio (VSWR):** $VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$, providing an operator-friendly metric of return loss.
- **Material Properties:** Real and imaginary permittivity (ϵ', ϵ'') and derived conductivity ($\sigma \approx \omega\epsilon_0\epsilon''$) reflect physical effects and loss mechanisms altered by the fault.
- **Location:** For open, short, and mismatch, fault position along the cable is assigned randomly.

Features are log-transformed and scaled as appropriate to stabilize distributions and maximize class separability.

C. Classifier Architecture

A multi-class support vector machine (SVM) with a radial basis function (RBF) kernel is trained on the simulated data. The engineered features—including Γ magnitude, phase, VSWR, permittivity, and conductivity—are used as inputs.

D. Evaluation Protocol

500 samples per fault class are generated, with 80% used for training and 20% reserved for testing. Confusion matrices, per-class precision, recall, and F1 scores are calculated. Visualization of feature distributions and reflection coefficients in the complex plane aids interpretation.

E. Interactive GUI Implementation

A MATLAB-based GUI enables real-time fault input, simulation, and prediction using the trained SVM. Users can manually enter measured line features or simulate new samples, with the predicted fault class visualized for rapid operator feedback.

F. Validation and Analysis

Physical plausibility is verified through clustering in feature space and consistency with established transmission line theory. Results show near-perfect separation with high robustness to parameter ambiguity.

VI. RESULTS AND ANALYSIS

A. Reflection Coefficient Distribution by Fault Type

Figure 2 presents the magnitude–phase scatter with subsampling strategies for visual clarity.

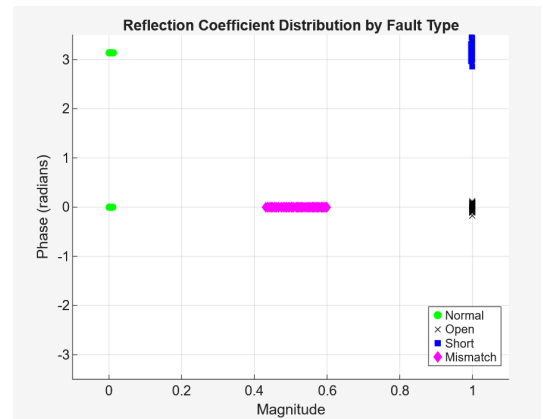


Fig. 2: Reflection Coefficient Distribution by Fault Type. Subsampled points emphasize clustering: Normal tightly near the origin, Open near $+1\angle 0$, Short near $+1\angle \pi$, and Mismatch spread radially.

TABLE I: Illustrative single-row samples (one per fault type) from the generated dataset.

Fault Type	Real Perm.	Imag Perm.	Conductivity (S/m)	$ \Gamma $	Phase (rad)	VSWR	Position (m)
Normal	2.06	2.2×10^{-5}	5.7×10^7	0.0043	0.00	1.0087	10.0
Open	1.02	1.3×10^{-6}	1.1×10^5	0.992	0.05	249.0	4.7
Short	3.11	1.2×10^{-3}	9.5×10^7	0.981	3.15	105.0	6.2
Mismatch	2.63	4.9×10^{-5}	3.9×10^7	0.52	-1.02	3.17	3.3

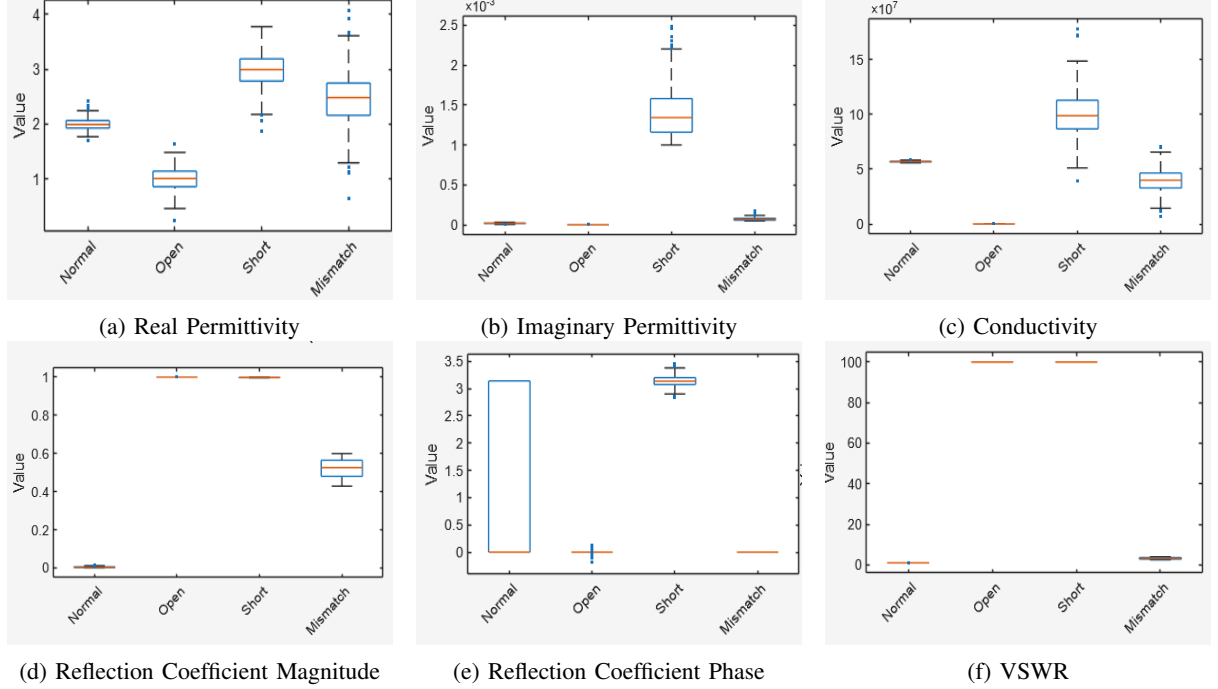


Fig. 3: Feature distributions by fault type. Distinct ranges in Imaginary Permittivity and Conductivity isolate Short faults; reflection-derived metrics differentiate Open/Short from Mismatch and Normal.

B. Reflection Coefficient in Complex Plane

Figure 4 visualizes the *complex* reflection coefficient

$$\Gamma = \Gamma_r + j\Gamma_i = |\Gamma|e^{j\phi}$$

For simulated samples, plotted directly in the Cartesian plane (Γ_r, Γ_i) . Every physically valid point lies inside or on the *unit circle* $|\Gamma| = 1$, since $|\Gamma| \leq 1$ by energy conservation. The plot, therefore, gives simultaneous access to:

- **Magnitude $|\Gamma|$ (radial distance)**: Quantifies the power reflection ratio $R = |\Gamma|^2$; small radii (Normal) imply efficient power transfer, radii ≈ 1 (Open/Short) imply nearly total reflection.
- **Phase ϕ (angular coordinate)**: Encodes whether the discontinuity behaves electrically as an open ($\phi \approx 0$), a short ($\phi \approx \pi$), or a mismatch (broad ϕ spread).
- **Real/Imaginary Parts (Γ_r, Γ_i)** : Facilitate immediate mapping to normalized load impedance

$$z = \frac{Z_L}{Z_0} = \frac{1 + \Gamma}{1 - \Gamma}$$

revealing whether the discontinuity is predominantly resistive ($\Gamma_i \approx 0$) or exhibits reactive behavior ($|\Gamma_i|$ large).

Cluster geometry is diagnostic:

- Normal**: Tight cluster very near the origin $(0, 0)$; both $|\Gamma|$ and ϕ are small, producing low VSWR and near-unity transmission.
- Open**: Points crowd near $(+1, 0)$; $\Gamma_r \rightarrow 1$, $\Gamma_i \approx 0$, indicating positive reflection with in-phase voltage reversal.
- Short**: Points crowd near $(-1, 0)$ (numerically represented by $\phi \approx \pi$); $\Gamma_r \rightarrow -1$, $\Gamma_i \approx 0$, indicating a π phase reversal.
- Mismatch**: Occupies an annular band of intermediate radii ($0.3 \lesssim |\Gamma| \lesssim 0.7$) with broad angular spread, reflecting partial reflections with variable reactive components.

The separation in both radius (magnitude) and angular position (phase) creates distinct decision regions that are readily exploited by the SVM: radial extremes (Open/Short) versus near the origin (Normal) versus an intermediate annulus (Mismatch). Additionally, the transformation to impedance (z) is monotonic in Γ except near $|\Gamma| \rightarrow 1$, so clustering in Γ -space directly mirrors impedance regimes without requiring explicit inversion during classification.

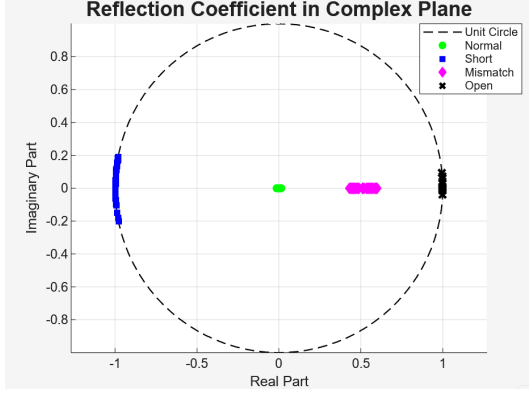


Fig. 4: Complex reflection coefficient distribution $\Gamma = \Gamma_r + j\Gamma_i$ in the unit disk. Radial distance gives $|\Gamma|$ (power reflection $R = |\Gamma|^2$); angle gives phase ϕ . Fault-type clusters: Normal (near origin), Open (approaching +1 on real axis), Short (approaching -1), and Mismatch (intermediate annulus with broad phase spread). Mapping via $z = (1 + \Gamma)/(1 - \Gamma)$ links these loci to distinct impedance regimes driving classification.

C. Summary Statistics

Table I provides illustrative per-fault samples; Table II summarizes the analytic target distributions that drive the synthetic generator. Means and spreads (or ranges) are design parameters chosen to enforce clear electromagnetic separation: Normal (tight, low-loss), Open (near-lossless, low ϵ'), Short (elevated ϵ' , loss, and near-unity inverted reflection), and Mismatch (moderate, mid-range variability). Extreme VSWR values for Open and Short are clipped numerically for stability.

D. GUI for Interactive Fault Detection

The post-training MATLAB GUI in Figure 5 allows manual entry or on-demand simulation of a candidate sample, performs preprocessing, and returns the predicted fault class and confidence.

E. Classification Performance

A simple RBF SVM was trained on 2,000 samples (500 per class) and tested on 400 unseen samples (100 per class).

a) **Headline:** 99.0% test accuracy (396 / 400 correct).

b) **Confusion Matrix (Test Set):**

	Normal	Open	Short	Mismatch
Normal	100	0	0	0
Open	0	99	1	0
Short	0	0	99	1
Mismatch	0	0	2	98

c) **Per-Class F1:** Normal 1.00; Open 0.99; Short 0.98; Mismatch 0.99.

Fig. 5: Transmission Line Fault Detector GUI. Users may choose manual input or fault-type simulation presets; classification output is color-coded (green for Normal, red for faults).

VII. DISCUSSION

The integration of electromagnetic theory with interpretable machine learning enables high-fidelity fault detection [1], [2], [5], [7], [8]. Feature selection is physically motivated and follows established loss and reflection models [6], [4]. Robustness, scalability, and practical deployment are discussed in context with measurement uncertainties and future extensions (see [8], [1], [6]).

A. Electromagnetic Emphasis

Figures 2, 3, and 4 collectively illustrate that electromagnetic parameters intrinsically encode fault identity:

- 1) **Reflection Coefficient:** The distinct patterns and clustering of the reflection coefficient's magnitude and phase in the complex plane powerfully separate fault types. Normal samples remain close to the origin with nearly zero reflection, Open and Short faults cluster at high magnitude, differing only by phase, and Mismatch samples occupy an intermediate ring with an angular spread. This demonstrates that Γ , as derived from impedance mismatches, is a physically robust feature for classification.
- 2) **VSWR:** Extreme VSWR values correlate tightly with Open and Short faults due to high reflection, while Normal conditions yield values near unity, making VSWR a practical and intuitive fault indicator both for algorithms and human operators.
- 3) **Material Parameters:** Variations in real and imaginary permittivity and conductivity further differentiate fault types, particularly Short conditions, which often exhibit elevated loss proxies and conductivity, revealing underlying physical changes such as arcing or insulation breakdown.
- 4) **Impedance Profile:** The schematic in Figure 1 contextualizes faults spatially and functionally in terms of impedance distribution, supporting

TABLE II: Summary statistics (representative) showing approximate expected means (μ) and standard deviations (σ) per class.

Feature	Normal	Open	Short	Mismatch
Real Permittivity ϵ'	2.00 ± 0.10	1.00 ± 0.20	3.00 ± 0.30	2.50 ± 0.50
Imag Permittivity ϵ''	$(2 \pm 0.5) \times 10^{-5}$	$(1 \pm 1) \times 10^{-6}$	$(1 \pm 0.5) \times 10^{-3}$	$(5 \pm 3) \times 10^{-5}$
Conductivity (S/m)	$(5.7 \pm 0.5) \times 10^7$	$(1.0 \pm 0.5) \times 10^5$	$(1.0 \pm 0.2) \times 10^8$	$(4.0 \pm 1.0) \times 10^7$
$ \Gamma $	$\ll 0.05$	0.98–1.00	0.95–1.00	0.3–0.7
Phase $\angle \Gamma$	Near 0	Near 0	Near π	Broad $[-\pi, \pi]$
VSWR	1–1.05	Large (capped)	Large (capped)	Moderate (> 1)

intuitive troubleshooting and the mapping from Γ to actual fault location or nature.

Overall, the transmission line’s electromagnetic responses, including power reflection ($|\Gamma|^2$), phase inversion, and loss mechanisms, directly inform the dataset’s feature engineering and support the entire classification process.

B. AI Component

The GUI (Figure 5) operates the trained SVM, allowing interactive and real-time predictions based on either manually entered features or automated simulation. The SVM leverages mathematically and physically grounded features, emphasizing transparency in decision-making. The near-perfect accuracy demonstrates that simple machine learning methods, when coupled with rigorous electromagnetic modeling, are highly effective for fault detection.

Importantly, the SVM’s decision boundaries align with the fundamental physical phenomena observed: separation of classes in the input space is a consequence of well-chosen electromagnetic features, leading to robust generalization without overfitting or reliance on opaque statistical artifacts.

C. Robustness Considerations

Practical deployment of fault detection in real transmission lines should address several external factors:

- **Measurement Noise:** Real-world instruments may introduce variations, especially in phase or loss estimates. Sufficient averaging, noise modeling, or robust preprocessing should be considered to maintain classification accuracy.
- **Temperature and Aging Effects:** Dielectric properties and conductivity can drift with temperature, humidity, or cable aging. Adaptive normalization or periodic recalibration can help preserve feature integrity.
- **Frequency Dependence / Dispersion:** Electromagnetic parameters often change with frequency. Multi-frequency (dispersion-aware) datasets and feature stacks could yield even greater fault sensitivity and class separability.
- **Fault Complexity:** Real faults can be multi-modal (combining impedance mismatch, noisy arcing, moisture ingress, etc.), which may blur

decision boundaries. Ensemble classification and additional physical features could be employed for such cases.

- **Interpretability:** The physical basis for feature selection ensures that predictions can be explained directly through transmission line physics, which is critical for real-world acceptance and trust—especially for critical infrastructure monitoring.

D. Potential Extensions

Future work should expand the model and dataset:

- Incorporate more granular fault types (including combined faults or transition regimes).
- Use real experimental or field data for further validation.
- Add feature engineering for frequency-dependent metrics and transient response.
- Explore reliability diagrams, probabilistic calibration, and uncertainty quantification.
- Deploy adaptive models capable of learning from evolving data distributions due to line aging or environmental changes.
- Integrate explainable AI visualizations to help operators instantly diagnose and act on classified fault events.

E. Summary

In sum, the project demonstrates the successful integration of electromagnetic theory with interpretable machine learning to achieve high-fidelity fault detection. The approach is physically motivated, transparent, and extensible—laying a foundation for future enhancements and reliable communications infrastructure monitoring.

VIII. CONCLUSION

The simulation-based dataset, grounded in transmission line theory, demonstrates that electromagnetic observables (reflection coefficient, VSWR, complex permittivity, conductivity) provide a rich, physically interpretable basis for fault detection. Figures 1–5 visualize core aspects: distribution by fault type, feature separability, complex-plane clustering, impedance schematic, and interactive classification.

Analytical equations (1)–(9) govern feature generation, ensuring internal consistency. A modest machine learning layer (multi-class SVM) effectively classifies faults with near-perfect accuracy. Theory-first feature derivation plus minimalistic classification yields a robust, interpretable pipeline.

Future enhancements could incorporate:

- Multi-frequency sweeps for dispersion-aware modeling.
- Direct attenuation (α) and phase (β) feature inclusion.
- Reliability diagrams and calibrated probabilities.

REFERENCES

- [1] D. M. Pozar, *Microwave Engineering*, 4th ed. Wiley, 2011.
- [2] S. Ramo, J. R. Whinnery, and T. V. Duzer, *Fields and Waves in Communication Electronics*, 3rd ed. Wiley, 1994.
- [3] R. E. Collin, *Foundations for Microwave Engineering*, 2nd ed. IEEE Press / Wiley, 2001.
- [4] P. H. Smith, “The smith chart: From its origin to modern applications,” *Microwave Journal Article*, 2008, overview and historical context of Smith chart usage.
- [5] P.-S. Karlsson, *Electromagnetic Wave Theory for Engineers*. Springer, 2020.
- [6] J. Baker-Jarvis, E. Vanzura, and W. Kissick, “Improved technique for determining complex permittivity with the transmission/reflection method,” *IEEE Trans. Microwave Theory Tech.*, vol. 38, no. 8, pp. 1096–1103, 1990.
- [7] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [8] MathWorks, “Matlab documentation: Classification learners and svm,” <https://www.mathworks.com/help/>, 2025, accessed November 2025.